

4.3.3 Handling Missing Values (NaN)

خالی خانے NaN ہیں۔

CS XII

Chapter 4: Data and Analysis

Lecture 05 of 9

★ Golden

4.3.3 Handling Missing Values (NaN) ★ Golden

LEARN IT

Missing cells are **NaN** ("Not a Number"). Statistics crash or lie if you ignore them. Three tools: `isnull()` (True where empty), `fillna(value)` (replace), `dropna()` (delete the row).

```
import pandas as pd
df = pd.DataFrame({
    "Name": ["Ali", "Sara", "Ahmed", "Ayesha"],
    "Marks": [85.0, None, 78.0, 92.0],
    "Grade": ["A", "A", None, "A"],
})
print(df.isnull().sum())
```

OUTPUT

```
Name      0
Marks     1
Grade     1
dtype: int64
```

WORKED EXAMPLE

Replace Marks with the mean of the known marks, and Grade with a label. Known marks 85, 78, 92 → mean = $255/3 = 85.0$.

```
avg = df["Marks"].mean()          # 85.0
df["Marks"] = df["Marks"].fillna(avg)
df["Grade"] = df["Grade"].fillna("Not Assigned")
print(df)
```

OUTPUT

	Name	Marks	Grade
0	Ali	85.0	A
1	Sara	85.0	A
2	Ahmed	78.0	Not Assigned
3	Ayesha	92.0	A

`df = df.dropna()` would instead delete Sara's row and Ahmed's row (any row that still had a NaN).

EXAM TIP

State: identify with `isnull / sum`, fill numeric gaps with the mean, fill categories with a label, or drop rows if filling would invent a lie. Never analyse first and clean second.

COMMON MISTAKE

Filling a salary gap with the mean of a company that includes the CEO can hide poverty (the same outlier problem as the mean). Know *why* the value is missing before you fill.

اردو میں سمجھیں

خالی خانے NaN ہیں۔ `isnull` تلاش، `fillna` بھرنا (اکثر اوسط سے)، `dropna` قطار ہٹانا۔ پہلے صاف کرو، پھر تجزیہ۔

CHECK YOURSELF

1. What does `df.isnull().sum()` return?
2. Does `dropna()` delete a column or a row by default?

Answers: 1. Count of NaNs in each column. 2. Any **row** that contains at least one NaN.

Next lecture: 4.4 Data Organization and Representation